Lecture 03.02: Practice Examples (Conditionals)

📌 Executive Summary

Real-world decision logic examples: Driving license eligibility, academic grading scales, and tax bracket calculations.

🔬 Theoretical Foundations & Principles

1. Complex Real-World Predicates

Combining multiple operational thresholds requires structured chained comparisons and range evaluations.

2. Multi-Tier Grading Bracket Model

Given numerical grade $G \in [0, 100]$:

$$\text{Grade}(G) = \begin{cases} 'A' & \text{if } 90 \leq G \leq 100 \\ 'B' & \text{if } 80 \leq G < 90 \\ 'C' & \text{if } 70 \leq G < 80 \\ 'D' & \text{if } 60 \leq G < 70 \\ 'F' & \text{otherwise} \end{cases}$$

3. Short-Circuit Evaluation Efficiency

When testing composite conditions with `and` / `or`, place cheap, highly selective conditions first to minimize expensive sub-expression evaluations.

💻 Production Implementation & Code

```
def calculate_grade(marks: float) -> str:
    if not (0 <= marks <= 100):
        raise ValueError(f"Invalid marks {marks}: must be between 0 and 100")
    if marks >= 90:
        return 'A'
```

```
    elif marks >= 80:
        return 'B'
    elif marks >= 70:
        return 'C'
    elif marks >= 60:
        return 'D'
    else:
        return 'F'

for score in [95, 82, 74, 61, 45]:
    print(f"Marks: {score} -> Grade: {calculate_grade(score)}")
```

⚙️ Algorithmic Complexity & Systems Analysis

- **Time Complexity:** Detailed in respective algorithmic operations.
- **Auxiliary Space:** Memory bounds and CPython frame management.